

System Specification

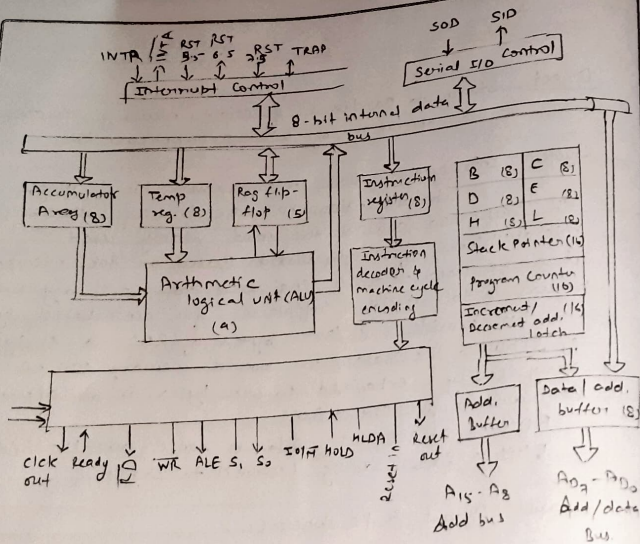
CPU	8 bit Microprocessor, the 8085
MEMORY	Total on board capacity of 64K bytes
RAM	8K/32K bytes & spaces for further expansion
ROM	8K bytes of EPROM loaded with prewritten program
TIMER	16 BIT PROGRAMMABLE timer/counter using 8255
8253 I/O	24 I/O lines using 8255
Keyboard	10 keys for command 16 keys for hexadecimal data entry 1 key for vector interrupt & 1 key for reset
BUS	All data address & control signal (TTL compatible available at FPC connector)
LED Display	6 Seven segment display & 4 for address field & 1 for data field
INTERFACE	RS-232-C through SID/SOD lines with auto baud rate
Power Supply	+5V, 1.5 Amp for the kit
Requirements	$\pm 12V$, 250 mA for CRT/DC interface
Operating Temp.	0 to 50°C

Object Study to hardware, functions, memory structure & operations of 8085 microprocessor kit.

System Introduction 8085 microprocessor kit is single board microprocessor training / development kit configured around the most widely used microprocessor of today world. Based on 8085 microprocessor it can be used to train engineers to control any industrial process & to develop software for 8085 based system. Kit provides 8K/32K bytes of RAM & 8K bytes of EPROM. The total on board memory can be very easily expanded to 64K bytes in an appropriate combination of RAM & ROM.
 ROM $\Rightarrow$ 8K EPROM (Address range - 0000H - 1FFFFH)
 RAM $\Rightarrow$ 8K range (2000H to 3FFFFH)

Input / Output & Peripherals.

- 24 programmable I/O lines using 8255
- 16-bit programmable Timer / Counter using 8253.
- RS-232-C interface through SID/SOD lines with auto baud rate.
- All address, data, control signals are TTL compatible & available via FPC connector.



Architecture of 8085

Hardware :-

The system is based on the 8085 microprocessor operating at 3.07 MHz, generated from a 6.14 MHz crystal oscillator.

The 8085 has -

8 data lines

16 address lines

The lower 8 address lines are multiplexed with the 8 data lines. During the first clock cycle, the lower 8 bits of the address appear on the bus, and in the next cycles, data appears.

To retain the lower address bits, a 74LS373 latch is used, ensuring the full 16-bit address remains available during operation.

Memory :-

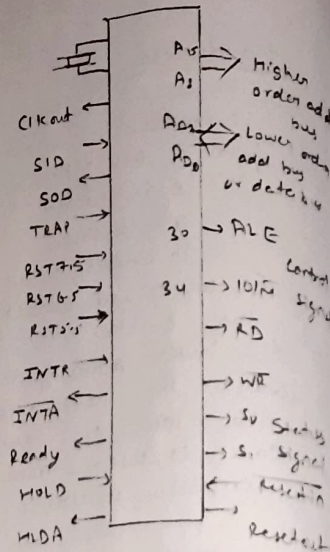
- RAM - 8K/32K using 6264/62256 chips
- EPROM - 8K for monitor program.
- One additional memory space (8000H - DFFFH) configurable based on chip size.
- Total onboard memory expandable up to 64KB.

I/O Devices :-

- 8255 - Keyboard & Display Controller
- Interfaced 28 key matrix keyboard & six 7-segment displays.
- Contains 16 x 8 display RAM.
- Provides FIFO for keyboard data.
- Automatically refresh display.

X ₁	1	40	VCC
X ₂	2	39	HOLD
Reset out	3	38	HLDA
SOD	4	27	CLC (out)
SID	5	26	Rektin
Trap	6	36	Ready
RST 7.5	7	34	IO/M
RST 6.5	8	33	S ₁
RST 5.5	9	32	FD
INTR	10	31	WR
INTA	11	30	ALE
AD ₀	12	29	S ₀
AD ₁	13	28	A ₁₅
AD ₂	14	27	A ₁₄
AD ₃	15	26	A ₁₃
AD ₄	16	25	A ₁₂
AD ₅	17	24	A ₁₁
AD ₆	18	23	A ₁₀
AD ₇	19	22	A ₉
AD ₈	20	21	A ₈

8085 Pinout



8255 - Programmable Princip Peripheral Interface (PPi)

- Provides 24 programmable I/O lines.
- Three 8-bit ports: Port A, Port B, Port C.
- Port C can be divided into upper & lower 4 bit.
- I/O configuration controlled by software.

8253 - Programmable Interval timer.

- Three independent 16-bit counters.
- Used for times delays, rate generation, RTC etc.
- Counter 0: Single-stop operation (1.535 MHz clock).
- Counter 1: Generates clock for 8251.
- Counter 2: Unavailable.

Display:- Six-7 segment hexadecimal displays.
4 digits → Address
2 digits → Data.

Command Description:-

The kit uses a 28-key board & 6-digit display.

- RESET: Resets system.
- VCT INT: Generates RST 7.5 interrupt.
- SHIFT: Enables secondary functions.
- INO: Executes / modify registers.
- O-F: Hexadecimal key.

Operation Notes:-

- All inputs are in hexadecimal.
- "-" on MSD → waiting for command.

"-Err" → invalid command.

Dot on LSD (address field) → system expects address

Dot on LSD (data field) → data modified
Commands use PREV, NEXT, and '.' (Execute) as delimiters.

Viva Question

Q1 In 8085 _____ interrupt has the highest priority.
(a) TRAP (b) INTR (c) RST 7.5 (d) RST 5.5

Ans (a) TRAP

Q2 The width of Data bus of 8085 microprocessor is
(a) 16 (b) 14 (c) 8 (d) 20

Ans (c) 8

Q3 In 8085 microprocessor _____ flag bit is internally used for BCD addition.

(a) Z (b) P (c) AC (d) C

Ans (c) AC (Auxiliary Carry flag)

Q4 The Microprocessor 8085 is a _____ pin IC.
(a) 80 (b) 20 (c) 120 (d) 40

Ans (d) 40

Q5 _____ signal is used to store lower 8-bits of address information in external latch.

(a) ALE (b) Ready (c) HOLD (d) RESETOUT

Ans (a) ALE (Address latch enable)

Teacher's Signature _____

Memory Location	Mnemonic	operand	opcode	Comments
C000H	LXI	H, C500	21	Load the memory address in HL pair, memory pointer.
C001H			00	
C002H			C5	
C003H	MOV	A, M	7E	Copy content of M to A
C004H	INX	H	23	Increment HL pair
C005H	MOV	B, M	46	Copy contents of M to B
C006H	ADD	B	80	Add contents of A & B $A + B \rightarrow A$
C007H	INX	H	23	Increment HL pair
C008H	MOV	M, A	77	Copy content of A to M; result stored in memory address.
C009H	HLT		76	Halt the program.

For addition of Two numbers.

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Object 1:- Add the contents of memory locations xx00 to xx01 to place the result in memory location xx02.

Apparatus Required - +5 Volt SMPS, Microprocessor 8085 Dyna-kit, Connecting wires.

Program Analysis:- For adding two bit numbers in microprocessor first we have to store the bit into the given address.

Sample problem:-

⑧ C500H = 07

C501H = 04

Result = 0B

The 8085 microprocessor is an 8 bit processor with 8 bit ALU.

To add two numbers into the accumulator.

1. Load first number into accumulator.
2. Add the second number to accumulator.
3. Store the result in another memory location.

- The LDA instruction loads data from a memory location into the accumulator.
- The ADD instruction adds register content to the accumulator.
- The STA instruction stores accumulator content into a memory location.

Teacher's Signature _____

output
A = 20H
memory location
205H = 20H

Result
~~2050 = 07H~~
Accumulator (A) = 07H
Register B = 02H
Memory location 2050H
= 07H

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Observation

INPUT Values.

C500 = 04

C501 = 07

Output Values

C502 = 0B

Code

MVI A, 20

STA 2005

HLT

Code for Sum of 2 numbers.

MVI A, 05H

MVI B, 02H

ADD B

STA 2050

HLT

Result:- Thus, we have written an assembly language program for addition & subtraction of two 8-bit numbers & stored in memory location.

Viva Question

Q1 What is the memory capacity of 8085 microprocessor?
Ans 64Kb

Q2 What is the function of timing & control unit?
Ans To control the timing operation of microprocessor.

Q3 What is the function of ALE signal?
Ans Address latch enable (ALE) signal is used to multiplexed and demultiplexed of Address & Data Bus ($AD_0 - AD_7$)

Q4 What is the function of READY signal?
Ans It is used by microprocessor to sense whether a peripheral is ready to interface or not for data transfer.

Q5 What do you mean by interrupt?
Ans The interrupt is defined as a signal which is generated by peripheral to break the sequence of the main program & program execution jumps to the pre-specified memory location.